INSTITUTE OF MATHEMATICAL SCIENCES
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SIT1001

Probability & Statistics I

TUTORIAL 1

1. An insurance company looks at its auto insurance customers and finds that (a) all insure at least one car, (b) 85% insure more than one car, (c) 23% insure a sports car, and (d) 17% insure more than one car, including a sports car. Find the probability that a customer selected at random insures exactly one car and it is not a sports car.

2. A coin is tossed four times, and the sequence of heads and tails is observed.
   a. List each of the 16 sequences in the sample space S.
   b. Let events A, B, C, and D be given by A = {at least 3 heads}, B = {at most 2 heads}, C = {heads on the third toss}, and D = {1 head and 3 tails}. If the probability set function assigns $1/16$ to each outcome in the sample space, find (i) $P(A)$, (ii) $P(A \cap B)$, (iii) $P(B)$, (iv) $P(A \cap C)$, (v) $P(D)$, (vi) $P(A \cup C)$, and (vii) $P(B \cap D)$.

3. Let p_n, n = 0,1,2, ... , be the probability that an automobile policyholder will file for n claims in a five-year period. The actuary involved makes the assumption that $p_{n+1} = (1/4)p_n$. What is the probability that the holder will file two or more claims during this period?

4. A boy found a bicycle lock for which the combination was unknown. The correct combination is a four-digit number, $d_1d_2d_3d_4$, where d_i, i = 1,2,3,4, is selected from 1, 2, 3, 4, 5, 6, 7, and 8. How many different lock combinations are possible with such a lock?

5. How many four-letter code words are possible using the letters in HOPE if
   a. The letters may not be repeated?
   b. The letters may be repeated?

6. The "eating club" is hosting a make-your-own sundae at which the following are provided:

Ice Cream Flavour	Toppings
Chocolate	Caramel
Cookies-n-cream	Hot fudge
Strawberry	Marshmallow
Vanilla	M&M's
	Nuts
	Strawberries

   a. How many sundaes are possible using one flavour of ice cream and three different toppings?
   b. How many sundaes are possible using one flavour of ice cream and from zero to six toppings?

c. How many different combinations of flavours of three scoops of ice cream are possible if it is permissible to make all three scoops the same flavour?

7. The following table classifies 1456 people by their gender and by whether or not they favour a gun law. Compute the following probabilities if one of these 1456 persons is selected randomly:

	Male (S_1)	Female (S_2)	Totals
Favour (A_1)	392	649	1041
Oppose (A_2)	241	174	415
Totals	633	823	1456

a. $P(A_1)$,
b. $P(A_1|S_1)$,
c. $P(A_1|S_2)$,
d. Interpret your answers to parts (b) and (c).

8. Suppose that $P(A) = 0.7$, $P(B) = 0.5$, and $P([A \cup B]') = 0.1$.
a. Find $P(A \cap B)$.
b. Calculate $P(A|B)$.
c. Calculate $P(B|A)$.

9. Paper is often tested for "burst strength" and "tear strength." Say we classify these strengths as low, middle, and high. Then, after examining 100 pieces of paper, we find the following:

	Burst Strength		
Tear Strength	A_1 (Low)	A_2 (Middle)	A_3 (High)
B_1 (Low)	7	11	13
B_2 (Middle)	11	21	9
B_3 (High)	12	9	7

If we select one of the pieces at random, what are the probabilities that it has the following characteristics:

a. A_1,
b. $A_3 \cap B_2$,
c. $A_2 \cup B_3$,
d. A_1, given that it is B_2,
e. B_1, given that it is A_3?

10. Let $P(A) = 0.3$ and $P(B) = 0.6$.
a. Find $P(A \cup B)$ when A and B are independent.
b. Find $P(A|B)$ when A and B are mutually exclusive.

11. Show that if A, B, and C are mutually independent, then the following pairs of events are independent: A and $(B \cap C)$, A and $(B \cup C)$, A ' and $(B \cap C')$. Show also that A , B' and C' are mutually independent.

12. Flip an unbiased coin five independent times. Compute the probability of
    a. HHTHT.
    b. THHHT.
    c. HTHTH.
    d. Three heads occurring in the five trials.

13. Bean seeds from supplier A have an 85% germination rate and those from supplier B have a 75% germination rate. A seed-packaging company purchases 40% of its bean seeds from supplier A and 60% from supplier B and mixes these seeds together.
    a. Find the probability $P(G)$ that a seed selected at random from the mixed seeds will germinate.
    b. Given that a seed germinates, find the probability that the seed was purchased from supplier A.

14. Suppose we want to investigate the percentage of abused children in a certain population. To do this, doctors examine some of these children taken at random from that population. However, doctors are not perfect: They sometimes classify an abused child (A) as one not abused (ND) or they classify a non-abused child (N) as one that is abused (AD). Suppose these error rates are $P(ND|A)$ = 0.08 and $P(AD|N)$ = 0.05, respectively; thus, $P(AD|A)$ = 0.92 and $P(ND|N)$ = 0.95 are the probabilities of the correct decisions. Let us pretend that only 2 percent of all children are abused; that is, $P(A)$ = 0.02 and $P(N)$ = 0.98.
    a. Select a child at random. What is the probability that the doctor classifies this child as abused. That is, compute
    $P(AD) = P(A)P(AD|A) + P(N)P(AD|N)$.
    b. Given that the child is classified by the doctor as abused, compute $P(N|AD)$ and $P(A|AD)$.
    c. Also, compute $P(N|ND)$ and $P(A|ND)$.
    d. Are the probabilities in (b) and (c) alarming? This happens because the error rates of 0.08 and 0.05 are high relative to the fraction 0.02 of abused children in the population.

15. A test indicates the presence of a particular disease 90% of the time when the disease is present and the presence of the disease 2 % of the time when the disease is not present. If 0.5% of the population has the disease, calculate the conditional probability that a person selected at random has the disease if the test indicates the presence of the disease.